Where Liquid Networks Help, Hurt, and Fail to Train: A Pre-Registered Boundary Map

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Abstract

Liquid neural networks — continuous-time cores such as LTCs and their closed-form successor (CfC), wired with NCP/AutoNCP — are claimed to grant perception-and-control policies greater robustness under distribution shift. We subject that claim to a pre-registered measurement program spanning three task regimes at core-parameter parity and equal training budget: open-loop classification (companion P0, PASTIS), open-loop onset detection (companion E6), and closed-loop control (state-loop, LiquidFlight v1.0; vision-loop, this work). In the richest regime — a vision-in-the-loop fly-to-target twin with a perceptual-shift ladder — no CfC arm reaches our pre-registered nominal-capability precondition (mean $\geq$90% success at core parity of 27,648 $\pm$2%) on either allowed learning rate, so the robustness claim is *untestable under our precondition* and no gate arm was ever evaluated out of distribution (OOD) under the frozen criterion; the only OOD data is the pre-registered sanity calibration (Section 7). Beneath that boundary we measure a *trainability tax*: at identical budget, a GRU control reaches 100% while the wired CfC peaks at 92% (per-leg means up to 75.3%, a 43 pp seed spread) and the dense CfC caps at 65%. We attribute trainability to wiring, localize a continuous-time Δt implementation dependency, and contribute a reusable gate/precondition methodology with an arithmetic early-stopping rule. A negative result, pre-registered, maps where the mechanism can and cannot be measured.

1 Introduction

Continuous-time recurrent networks have become an attractive substrate for embodied perception and control. Liquid Time-Constant networks (LTCs) and their closed-form counterpart (CfC), especially when wired with Neural Circuit Policies (NCP/AutoNCP), have been reported to yield policies that generalize better under distribution shift than conventional gated recurrences such as GRUs. The most cited demonstration in the flight domain reports that a compact liquid policy attends to the task-relevant part of the visual scene and holds up under perceptual perturbation [Chahine et al., 2023].

Two properties of that literature motivate this work. First, demonstrations are typically reported **without core-parameter parity** against a matched baseline and **without pre-registration** of the success criterion; a favorable curve is shown, but the counterfactual — a same-budget, same-data baseline under a criterion fixed in advance — is often absent. Second, the field does not report a **trainability layer**: whether the liquid core can be brought to competence at all, at parity, under a fixed budget, before any robustness question is even asked.

We take the claim seriously enough to try to falsify it under conditions it should survive. We build an honest twin — identical encoder, identical head, identical data and budget, cores matched to within $\pm$2% — and we freeze the success criterion before measuring. This paper reports the vision-loop regime (phase 3) as the primary result and folds it into a portfolio-level **boundary map** with two companion regimes.

Contributions (mapped to our claim register C1–C8):

- A pre-registered vision-loop twin in which the robustness claim is *untestable under our precondition* — the deciding arm never qualifies, and we hold to zero OOD evaluations of any gate arm (C1).
- A measured **trainability tax** at parity: GRU 100% vs. wired-CfC (peak 92%, per-leg means up to 75.3%) vs. dense-CfC (cap 65%), at identical budget (C2).
- The finding that the boundary is **variance, not mean** — seed instability is the object of measurement, echoing the companion P0 population spread (C3).
- A **wiring attribution**: AutoNCP makes the CfC trainable where the dense CfC caps (C4).
- Engineering findings on the continuous-time Δt path and DAgger procedure (C5, C6).
- A transferable **methodology**: gates, preconditions, annexes with a stop rule, and arithmetic early resolution (C7); plus an instrument characterization (C8).

2 Related Work

Liquid networks. Liquid Time-Constant networks model the hidden state as a continuous-time dynamical system whose time constants are themselves input-dependent, giving the network a natively continuous notion of elapsed time [Hasani et al., 2021]. Their closed-form successor, the CfC, replaces the numerical ODE solve with a closed-form update that preserves the continuous-time gate while removing solver cost [Hasani et al., 2022]. Neural Circuit Policies and their automatic wiring (AutoNCP) impose a sparse, structured connectivity on the recurrent core, inspired by biological circuits and reported to improve auditability and compactness [Lechner et al., 2020]. The robustness claim we test derives largely from a flight-navigation demonstration in which a small wired liquid policy generalizes out of distribution and attends to task-relevant structure [Chahine et al., 2023]; we treat that claim as the hypothesis under test and its lineage as the object we reproduce at parity. Two of our arms — a wired CfC (A_NCP) and a dense CfC (A_CFC) — are chosen precisely to isolate the contribution of the wiring from that of the continuous dynamics.

Continuous-time and state-space cousins. Deep state-space models share the continuous-time framing: S4 parameterizes long-range dependencies through a structured state-space kernel [Gu et al., 2022], Mamba adds input-dependent selectivity for linear-time sequence modeling [Gu and Dao, 2023], and Liquid-S4 connects the liquid formulation to this family [Hasani et al., 2023]. These situate our small recurrences within a broader continuous-time lineage; we do not evaluate them here.

Pre-registration, reproducibility, and imitation learning. Our methodology adopts the discipline of freezing success criteria before measurement and reporting honest baselines, in the spirit of the machine-learning reproducibility and pre-registration literature [Pineau et al., 2021, Søgaard et al., 2023]. Training uses behavioral cloning followed by DAgger, the canonical no-regret reduction of imitation learning to online learning [Ross et al., 2011]; a central engineering finding of this work concerns the interaction between the DAgger update mode and continuous-time cores (Section 6). The vision harness is built on gym-pybullet-drones [Panerati et al., 2021], and the companion open-loop regime uses the PASTIS satellite time-series benchmark [Sainte Fare Garnot and Landrieu, 2021].

3 The Measurement Program

The program is a house methodology applied uniformly across regimes.

Honest twin. Each arm shares the same encoder, the same `Linear(64→6)` head with identical output scaling, the same training data and budget; cores are matched to a reference of 27,648 core parameters to within $\pm 2\%$. The only sanctioned inter-arm difference is the learning rate, whose provenance is documented.

Frozen criteria; MEASURE = REPORT. Success criteria are frozen in a gate document before any measurement; whatever is measured is reported, including nulls and boundaries. The vision-loop gate (F3-GATE) fixes the OOD level (T2b), the primary metric, the decision threshold, and n before any thesis measurement.

Gates and preconditions. A capability **precondition** (nominal success $\geq$90% per arm) must pass before the thesis metric is evaluated; the precondition is *not* the thesis. Failing it triggers a sanctioned, nominal-only repair (a learning rate from the pre-registered fallback grid — nothing else) and a full retrain, logged; exhaustion is a STOP for a human decision, not a verdict on the thesis.

Annexes with a stop rule. When the recipe diverges from a frozen, known-good reference, we unfreeze via an **annex** that cites the reference line-by-line and applies the change symmetrically to all arms. The stop rule (Annex-4) declares the last instrumental annex of phase 3a: if a CfC arm still fails the precondition afterward, FAIL becomes a reported **boundary**, not another annex.

Arithmetic early resolution. Because the precondition is a mean over 10 seeds, a leg can be resolved before all seeds run: after k seeds summing to S, if $S + (10 - k) \cdot 100 < 900$ the leg cannot reach 90% and is declared FAIL. This is arithmetic on the frozen criterion, not a change to it. In phase 3 it reduced the run to **13 full cycles instead of 40** and left one four-seed batch never launched (Appendix D).

Determinism contract. Environment, rendering, and seeding are bit-exact reproducible **within a single machine** (two runs, rgb/kin/setpoint identical). Cross-machine hashes may differ and that is not a failure — determinism is a within-machine property, not a cross-machine guarantee.

4 Harnesses

The portfolio uses three instruments, one per regime.

4.1 PASTIS twin (P0). Sentinel-2 crop classification of a phenologically confusable pair (soft winter wheat vs. corn) under test-time observation dropout, with a shared CNN encoder (28,752 parameters) and only the recurrent core swapped between CfC and GRU. Open-loop, day-scale. Companion; numbers verified against the companion report [Żydziak, 2026].

4.2 LiquidFlight (state-loop, v1.0). A closed-loop drone flight twin (gym-pybullet-drones [Panerati et al., 2021]) with a setpoint→DSL-PID inner loop at 48 Hz, an observation-dropout OOD axis and a gap-length (latency) axis, and a time-constant (τ) inspection panel; CfC(32) vs. GRU at core parity, everything but the core bit-identical between arms. The vision-loop harness reuses this execution layer verbatim (sha256-pinned copies).

4.3 LiquidSight (vision-loop, phase 3 — this work). A fly-to-target task from a $64 \times 64 \times 3$ forward camera plus kinematics: reach a ground marker and hold a hover within `r_goal` for the final dwell window. Physics at 240 Hz, control at 48 Hz, camera and policy tick at 12 Hz; the target lives only in pixels. A privileged DSL-PID expert supplies behavioral cloning (BC) labels; three DAgger rounds follow. The perceptual-shift axis is a distractor ladder T0–T3 (T2/T2a/T2b/T2c $= K \in \{0, 1, 2, 3\}$ distractors on structured backgrounds), with P-SANITY establishing that the axis resolves difficulty and that the expert ceiling is reachable at every level (Section 7). The gate level is T2b.

5 Results Across Regimes

The boundary map (Table 1) summarizes the direction and measurement resolution of the effect in each regime. No measurement crosses its pre-registered threshold.

Table 1: Boundary map across task regimes.

row	regime / harness	effect direction	verdict / resolution
R1	open-loop classification (P0)	weak positive under dropout	null (margin < pooled std); quantitative
R2	onset detection, low FAR (E6)	**negative**	directional; qualitative (secondary source)
R3	state-loop control (v1.0)	zero	quantitative
R4	vision-loop (this work)	untestable under precondition	precondition FAIL at parity; quantitative

5.1 Vision-loop (phase 3) — the primary result

The robustness claim is untestable under our precondition (C1). After restoring the known-good recipe in full (core construction, Annex-3; training procedure, Annex-4) and exhausting the only sanctioned lever — the learning-rate grid {3e-4, 1e-3} — **no CfC arm reaches the pre-registered precondition** of mean ≥90% nominal success at core parity. All four lr legs resolve to FAIL by arithmetic early resolution: A_NCP at 3e-4 = 16.5% (resolved at k=2), A_NCP at 1e-3 = **72.2%** (k=4), A_CFC at 3e-4 = 27.0% (k=2), A_CFC at 1e-3 = 55.3% (k=3). Because the **deciding arm** (A_NCP, the AutoNCP-wired CfC faithful to the wiring in the cited claim) does not qualify, the thesis metric — a T2b margin of A_NCP over A_GRU with threshold $M >$ pooled_std, n=10 — is *untestable under our precondition.* We therefore ran **zero OOD evaluations of any gate arm** in the entire phase: gate stages B (the A_GRU precondition sweep) and C (the OOD ladder) were skipped by the branch, so the pre-registered verdict remains untouched and unprejudiced. The only out-of-distribution data seen anywhere in the phase is the pre-registered sanity calibration of the perceptual-shift axis (the P-SANITY ladder), which is a pre-gate instrument check, not a thesis measurement (Section 7).

This is deliberately *not* a falsification of the thesis. A test requires a qualifying deciding arm; without the precondition there is no test. The result is a cell in the boundary map, not a verdict on the mechanism.

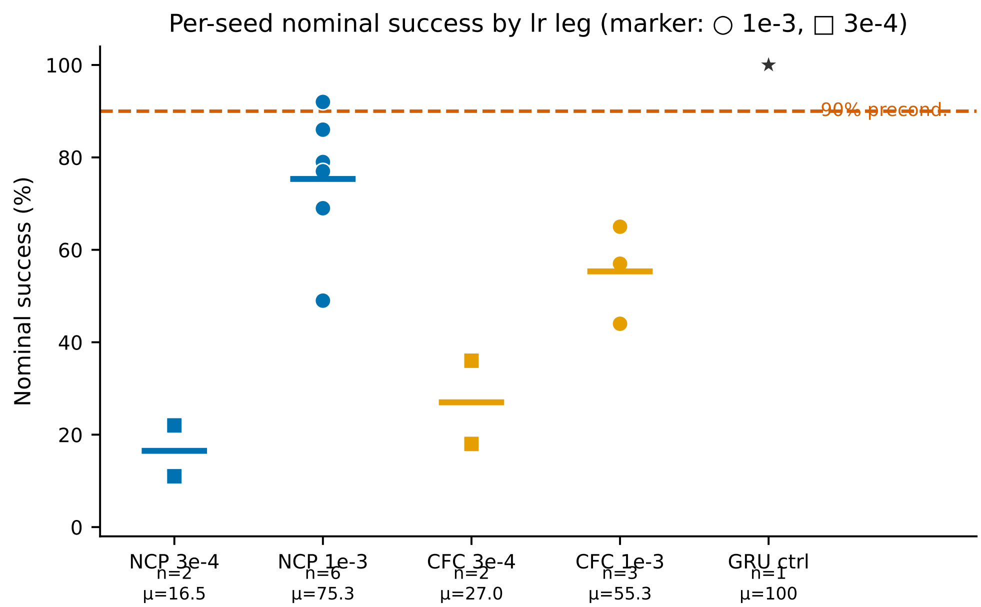

Figure 1: *Per-seed nominal success for each lr leg.* Points show nominal success (100 scenes, level T0) for every trained seed across the four CfC lr legs (A_NCP and A_CFC × {3e-4, 1e-3}); horizontal bars mark per-leg means; a dashed rule marks the 90% precondition. The A_GRU control (100%) is shown for reference. The wired arm at 1e-3 spans 49–92% (43 pp); no leg mean crosses 90%.

A trainability tax at parity (C2). At identical data, budget, and procedure, and with cores matched to ±2%, nominal capability forms a gradient: **A_GRU 100%** (stable; dwell failures 0; 100% in every GRU training of the program: P-SANITY, procedure-v2 smoke, and the binding control) > **wired CfC** (A_NCP; peak 92% at seed 45010/1e-3; per-leg mean 72.2% at k=4) > **dense CfC** (A_CFC; cap 65% at seed 45012/1e-3). Core-parameter parity alone does not buy capability; the continuous-time core pays a trainability tax at this budget. The dominant residual failure is **dwell** — the drone reaches the marker but does not hold the hover precisely — while catastrophic **tilt is eliminated** at 1e-3 (0 tilt across all CfC cycles, versus 78/100 for the dense CfC under the pre-Annex-4 procedure).

The boundary is variance, not mean (C3). The wired arm at 1e-3 spans **49–92% across seeds — a 43 pp spread** — under one fixed procedure, while the GRU control is a stable 100%. Reporting only the k=4 mean (72.2%) understates the phenomenon; the **full six-seed characterization (75.3%, spread 43 pp)** shows the boundary is a property of the between-seed distribution, not a sample-size artifact. The CfC at parity does not "fail to learn"; it learns **unstably and dwell-limited, on average below 90%**. This mirrors the companion P0 result [Żydziak, 2026], where the retention margin (+0.1127) and its pooled standard deviation (0.1759) are nearly invariant between n=3 and n=15 — population spread that does not shrink with n.

Wiring attribution (C4). The contrast between arms attributes trainability to the wiring: the AutoNCP-wired CfC reaches a 92% peak and per-leg means up to 75.3%, whereas the dense CfC caps at 65%, across seeds and at the 1e-3 lever. The attribution holds on the 1e-3 lever; at 3e-4 the ordering inverts (dense 27.0% > wired 16.5%), so the wiring benefit is itself learning-rate dependent and we report it as such. A historical, single-seed diagnostic corroborates the readout dimension: reading the full state rather than six motor neurons raises reach from 8/50 to 26/50 (Section 6).

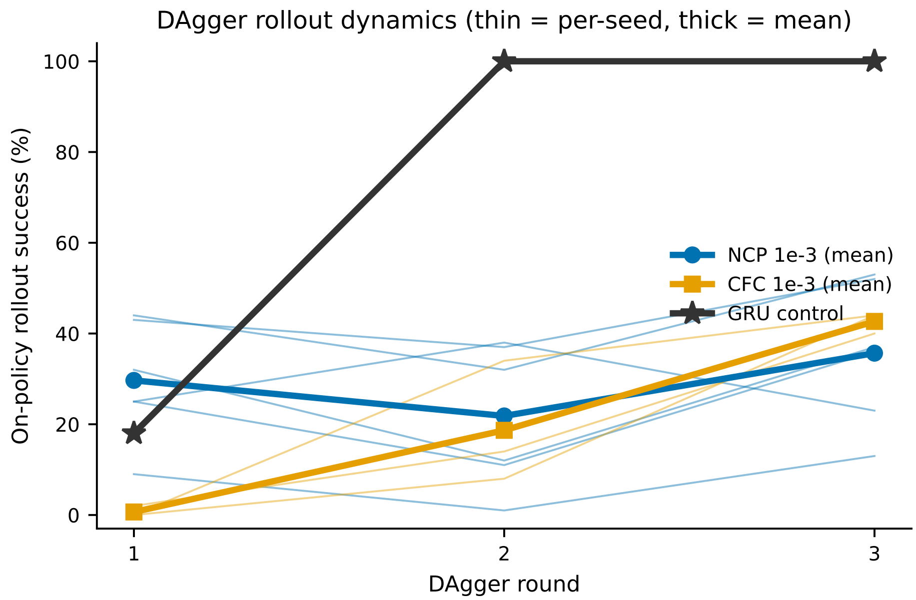

Figure 2: *DAgger rollout dynamics per round.* On-policy rollout success across DAgger rounds r1→r2→r3. The GRU control climbs 18→100→100; CfC arms stay low or flat, illustrating that the aggregate does not close for the continuous-time cores (thin: per-seed; thick: per-leg mean).

5.2 Open-loop classification, day-scale (P0) — companion

The PASTIS twin gives a directional-but-null signal: at full cadence the GRU leads on macro-F1 (0.9073 ±0.0137 vs. 0.8802 ±0.0121), but under observation dropout the CfC retains more — retention $R(0.6)$ of 0.6415 ±0.1138 vs. 0.5288 ±0.0622, a +0.1127 **margin against a pooled std of 0.1759, i.e. null** by the pre-registered rule. An absolute-F1 crossover appears at d=0.6 (CfC 0.5642 ±0.0985 above GRU 0.4798 ±0.0567, despite a −0.027 start), and the degradation slope is gentler for the CfC (−0.5095 vs. −0.6936). Crucially, margin and pooled std are near-invariant from n=3 to n=15 — the boundary is population spread, not sampling noise. All P0 numbers are verified against the companion report [Żydziak, 2026].

5.3 Open-loop onset detection, low FAR (E6) — companion

The onset-detection regime contributes the **negative direction** of the map: at a low false-alarm-rate operating point the effect sign is **unfavorable to the liquid core**, as recorded in the program compendium; the primary E6 report is not preserved in the project archive. We therefore state this regime **qualitatively only** and report no numerical values for it — in particular the detection-delay delta, the sweep parameter, and the false-alarm-rate definition are not carried into the prose, since their primary source could not be verified. The regime enters the boundary map as a directional cell (Table 1, row R2), not as a quantitative result.

5.4 State-loop control, millisecond-scale (LiquidFlight v1.0) — companion

In the state-loop regime the continuous-time lever (Δt) is not behaviorally load-bearing. With a setpoint action abstraction the CfC-32 policy flies under 500–1300 ms observation gaps and the episode-failure cliff moves from ≈102 ms (raw-RPM stabilization) to ≈779 ms; the Δt mechanism is *visible* in the state dynamics (recovery after a gap) but yields **no behavioral edge** — ablating the Δt channel flies identically. The fitted time constants are short (median $\tau \approx 35$ ms, IQR 24–69 ms), so a 500 ms gap is ≈14× τ and the model does not bridge gaps with long memory; τ serves as an explainability handle, not a source of advantage. Under the A1 perturbation grid there were **0 tip-overs**. The regime contributes a **zero** to the map: neither help nor harm attributable to the liquid core at this horizon.

6 Engineering Findings

Bringing a CfC to the point where it will run at all in this harness required restoring a known-good recipe in four provenanced steps (fixes **F1–F4**); each is an **engineering finding, established by single-seed controlled probes (diagnostic, not statistical)** — not a statistically powered ablation.

F1 — Backbone. The dense CfC built without a backbone under the parity constraint does not reach the target (reach 8/50); adding a frozen-style backbone restores it (35/50, ≈ GRU) (Annex-3).

F2 — The Δt path and its units (C5). The continuous-time core — the sole locus of the thesis mechanism — is load-bearing only after two corrections. First, the time-span scale sets the gate regime: with an explicit `ts` in **seconds** (0.0833 s per 12 Hz tick) reach is 39/50, versus 15/50 at ts=1.0 (ticks) and 9/50 at ts=4.0; the post-init gate drive $|t_a \cdot \mathrm{ts}|$ scales accordingly (0.010 / 0.125 / 0.501), so a mis-scaled `ts` freezes or destabilizes the cell. Second, we found a **library bug**: `ncps.torch.CfC` rejects explicit `timespans` when the batch size exceeds one (scalar, $(B, 1)$, and $(B,)$ all raise; only `None`=1.0 works), which had trapped the arms at ts=1.0. We work around it by stepping the cell manually. The auxiliary thesis is therefore: **Δt is load-bearing implementationally, not advantageously** — in this harness we could not measure an advantage because the precondition was never met.

F3 — Full-state readout. Reading the wired core's full state rather than six motor neurons lowers BC error (0.0047 vs. 0.0220) and raises reach (26/50 vs. 8/50) (Annex-3).

F4 — DAgger update mode (C6). Continuous-time cells are sensitive to the DAgger update procedure. Under the gate's original v1 mode (continue from the previous weights + final-epoch checkpoint), the dense CfC becomes unstable (tilt 78/100) and the wired CfC collapses its rollouts (1→0→0); the dense CfC is stable after BC alone (0 tilt), so the instability is **induced by the procedure**, not the construction. Adopting the frozen-C1 mode symmetrically (retrain from scratch each round + best-validation checkpoint, 120 epochs per stage; Annex-4) removes the tilt (78→0 at 1e-3) and follows the canonical DAgger formulation [Ross et al., 2011]. The **GRU is robust to both procedures** (100% either way).

A mechanistic signature accompanies this (C6a): for every CfC arm the best validation error after BC alone is low (≈0.00012–0.0009) but **rises** after the DAgger aggregate (rounds r1–r3), and the best epoch moves early (e.g., wired at 1e-3, seed 45013: 119→41→26→37) — the core does not close the fit to the policy-visited distribution as well as it fits the clean expert. The GRU holds its best validation low across all rounds (.000168→.000112) — a different regime.

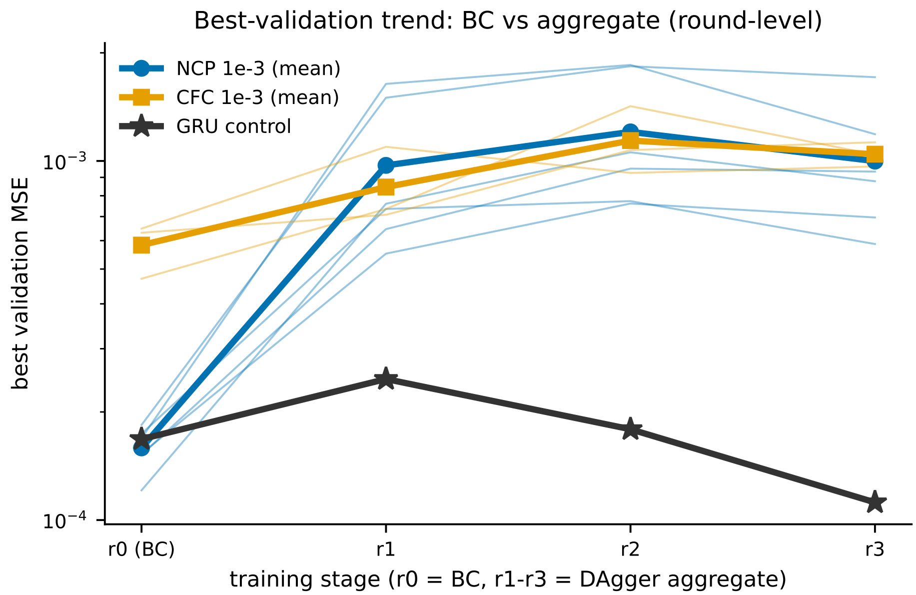

Figure 3: *Best-validation trend, BC vs. aggregate.* Best validation MSE per stage (r0 = BC → r1–r3 = DAgger aggregate) for the CfC arms (rising after aggregation, best epoch moving earlier) against the GRU control (monotonically low). Round-level (four points per cycle); full per-epoch curves are not persisted and would require re-measurement.

7 Instrument Characterization and Protocol Economics

Axis resolution and expert ceiling (C8). The perceptual-shift axis resolves difficulty monotonically. The sanity policy degrades **100/100/64/46/36/24/16** across T0/T1/T2/T2a/T2b/T2c/T3 (50 scenes per level), placing three levels inside the pre-registered [30%, 85%] band and selecting **T2b (36%)** as the hardest in-band gate level. The privileged expert reaches a **100% ceiling at every level**, confirming that the distractors do not block physical reachability — the axis touches pixels only. Determinism is bit-exact **within a machine** (rgb/kin/setpoint identical across two runs; cross-machine equality is neither required nor claimed).

Protocol economics (C7a). The binding run comprised **13 full training cycles** (plus one GRU control), ≈**57.1 h** of cumulative compute. Per-cycle wall times in the binding run are inflated by GPU contention (3–6 concurrent processes on one GPU). Solo-cycle costs were measured only for the GRU control and the 3e-4 legs (**1.7–2.4 h per cycle**); the 1e-3 CfC legs ran only under contention, so their solo cost is unmeasured. The 57.1 h figure is therefore a *sum of contention-inflated per-cycle times*, not a serial wall-clock (Table T7). Arithmetic early resolution spared a complete four-seed batch relative to the 40 cycles a full n=10 over four legs would have required.

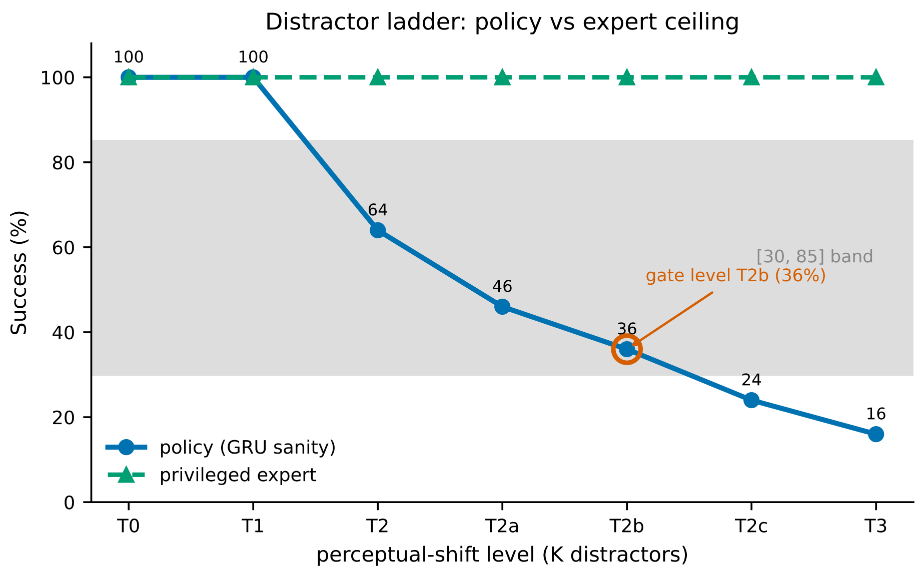

Figure 4: *Sanity ladder and expert ceiling.* Policy success across the distractor ladder (T0→T3: 100/100/64/46/36/24/16) with the [30,85] band shaded and the T2b gate level marked; the expert ceiling (100% at every level) overlaid.

8 Discussion

Synthesis. Across the regimes we map, the sign of the effect depends on the task: a weak positive in sparse open-loop observation (P0), a negative under low-FAR onset detection (E6), a zero in state-loop control (LiquidFlight), and — in the richest regime, vision-in-the-loop — a question that never reaches the conditions of measurement, because the advantage is preceded by a trainability tax at parity. No measurement crosses its pre-registered threshold. The unifying reading is that **task construction decides whether the mechanism can reveal itself at all**: the continuous-time core needs a regime whose dependencies it is positioned to exploit, and at parity it must first be trainable there.

Demonstration culture vs. falsification. A pre-registered null carries evidential weight that a single favorable curve does not. Much of the liquid-network robustness literature is demonstrated rather than falsified: a favorable model is shown on a task chosen after the fact, without a same-budget parity baseline and without a criterion fixed in advance. Our program inverts that stance — it commits to a threshold, a level, and a seed count before any out-of-distribution scene is seen, and it reports whatever the instrument returns, including a boundary. The cost of this discipline is that we sometimes conclude "untestable" rather than "wins" or "loses"; the benefit is that the three cells we *can* state (weak-positive, negative, zero) are not artifacts of selective reporting. The map says where to look and where not to — it does not claim the mechanism is absent, only that under parity, fixed budget, and frozen criteria we did not measure it in these harnesses.

When the twin re-arms. The vision-loop cell would re-enter measurement under conditions that unlock trainability without breaking parity. The cleanest such condition is a **symmetric budget increase** for all arms — more epochs, more DAgger rounds, or more data applied identically to GRU and CfC — carried until the deciding arm (A_NCP) meets the precondition; the wired arm at

1e-3 is the natural starting point, already at 92% best seed and 75.3% mean. Because the increase is symmetric, the duel remains fair: if the GRU also improves, the parity contract is preserved. Two subsidiary questions gate the value of that rerun. First, the sensitivity of continuous-time cells to the DAgger update mode (F4) is an open research problem worth isolating on its own — why does retrain-from-scratch with best-validation selection stabilize a CfC that continuation destabilizes, when a GRU is indifferent to both? Second, whether the residual dwell failure reflects a precision limit of the continuous-time state under this readout, or merely under-training, is answerable only once the precondition is met. Until then, we resist reading the boundary as evidence about the mechanism itself.

Next habitat. The mechanism's most plausible next habitat is a regime with genuinely variable time steps or asynchronous/event-stream sensing, where the continuous-time gate has something to integrate — offered as a direction, not a promise.

9 Limitations

- **Simulation only.** PyBullet dynamics, no photorealism, no sim-to-real transfer.
- **One harness per regime.** A single fly-to-target task family at 64×64; results may not generalize across task families.
- **Parity at ≈27k core parameters.** Findings are stated at this capacity; other capacities are untested.
- n=10 **with a finality rule.** Seeds 45010–45015 were used (early resolution); increasing n, or changing the level or threshold after seeing any OOD result, is forbidden by the gate.
- **No OOD measurement of any gate arm in the vision-loop regime.** This is a *consequence* of the unmet precondition, not a design choice; the only OOD data anywhere in the phase is the pre-gate sanity calibration.
- **Single-cycle GRU control.** The binding-run GRU control is a single cycle, corroborated by two earlier independent 100% cycles.
- **Engineering findings are single-seed probes.** The four recipe restorations (fixes F1–F4) are diagnostic controlled probes (n=1 for most), illustrative rather than statistically powered.
- **Determinism is within-machine**, not cross-machine.
- **Fixed training budget.** The trainability tax is measured at BC-120 + DAgger×3; a higher symmetric budget is unmeasured.
- **Single-operator program**; one task family per regime.
- **Companion E6 primary report not preserved.** The onset-detection regime (R2) is stated qualitatively from a secondary source; its primary report is absent from the archive, so R2 carries no numerical values and its direction should be read as directional evidence, not a measured effect size.

10 Reproducibility and Artifacts

The liquidflight v1.0 and liquidsight repositories (tags and gate commits), the frozen documents (F3_PRE0, decisions and Annexes 1–4, P-SANITY, F3-GATE), sha256 manifests of the execution layer, the seed pools, the per-cycle logs, and the reports **will be released upon publication**. The companion P0 ships as a separate PDF/arXiv artifact [Żydziak, 2026]. Instrument determinism is within-machine bit-exact; bit-for-bit reproduction across different GPUs is not guaranteed (inherent to CUDA).

11 Conclusion

We set out to test a robustness claim about liquid networks and, in the richest regime, found the claim **untestable under our pre-registered precondition**: at core parity and fixed budget, no continuous-time arm becomes reliably competent, so no out-of-distribution verdict can be earned. The first-order result is therefore a **boundary map** — where the mechanism helps (weakly), hurts, is neutral, and cannot yet be measured — together with a **trainability tax** that the field does not usually report: a same-budget capability gradient from a stable GRU (100%) through a wired CfC (peak 92%) to a dense CfC (cap 65%). The measurement methodology — gates, preconditions, provenanced annexes with a stop rule, and arithmetic early resolution — is the portable contribution. A negative result, pre-registered, is a map of where to measure next.

A Program timeline

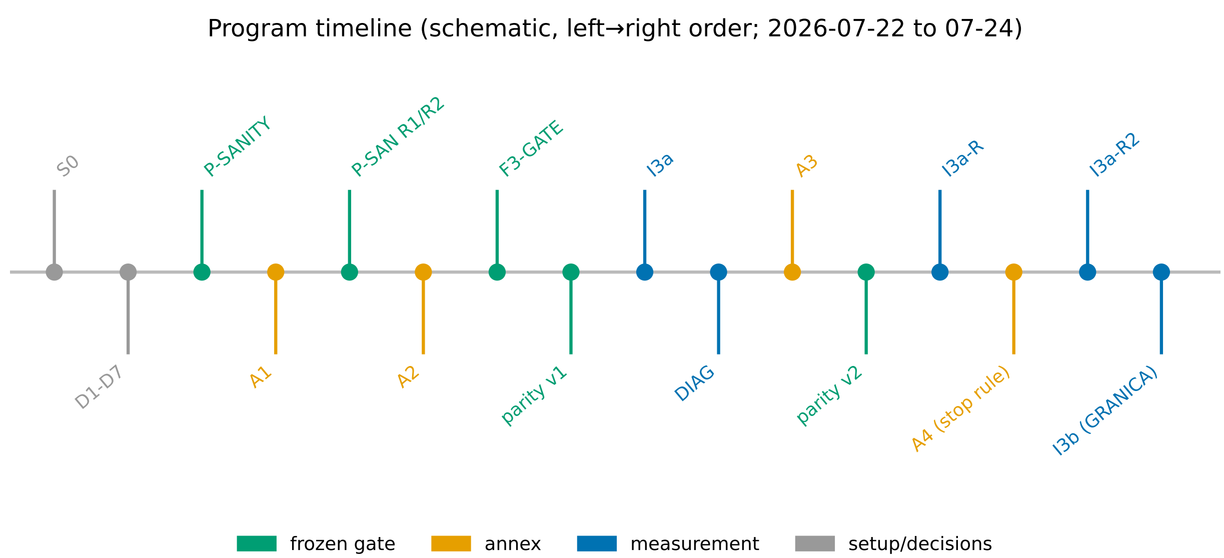

Figure 5: *Gate and annex timeline.* The phase-3 program as a left-to-right order (2026-07-22 to 07-24): frozen gates (P-SANITY, F3-GATE, parity v1/v2), annexes (A1 observability, A2 distractor ladder, A3 core construction, A4 training procedure), and measurements (I3a, DIAG, I3a-R, I3a-R2, I3b).

One-line reasons: **Annex-1** spawned the target in the frontal cone and pitched the camera −22.3° so the marker is in-frame; **Annex-2** extended the axis with T2a/T2b/T2c (K=1/2/3) to fill resolution; **Annex-3** fixed three CfC construction defects (backbone, ts-in-seconds, full-state readout); **Annex-4** unified the training procedure with frozen C1 (from-scratch per round, best-val, 120 epochs) and declared the stop rule.

B Frozen criteria (pointers)

- **P-SANITY**: P1 capability ≥90%; P2 axis resolution ≥2 levels in [30,85]; P3 expert ceiling ≥95%/level.
- **F3-GATE §4**: precondition mean nominal success ≥90% per arm; fallback grid {3e-4, 1e-3}; no OOD before all arms pass.
- **F3-GATE §5**: primary metric $M = \text{mean}_s\, \text{succ}(\text{A\_NCP}, s) - \text{mean}_s\, \text{succ}(\text{A\_GRU}, s)$, n=10; pooled_std $= \sqrt{(\text{sd}(\text{A\_NCP})^2 + \text{sd}(\text{A\_GRU})^2)/2}$; verdict $M >$ pooled_std; binding at n=10.
- **Stop rule** (Annex-4): the last instrumental annex of phase 3a; a subsequent precondition FAIL becomes a reported boundary.

C Per-seed tables (binding run I3b)

Columns: nominal % | failures | DAgger rollout r1→r2→r3 | best-val r0→r3 | best-epoch | cycle seconds. Generated from the per-cycle log (read, not measured).

Table 2: A_NCP @ 1e-3 (best leg; mean 72.2% at k=4, 75.3% over six seeds, 43 pp spread).

seed	nom%	failures	rollout	best-val r0→r3	sec
45010	92	dwell 8	44→32→53	.000174→.000587	18,889.3
45011	79	dwell 21	43→37→52	.000153→.000696	19,138.1
45012	69	dwell 31	25→11→36	.000169→.001187	19,441.2
45013	49	dwell 51	9→1→13	.000184→.001711	18,986.6
45014	86	dwell 14	32→12→37	.000121→.000878	19,471.4
45015	77	dwell 23	25→38→23	.000152→.000934	18,992.2

Table 3: Other legs and the GRU control (procedure v2).

leg	seed	nom%	failures	rollout	best-val r0→r3
A_NCP 3e-4	45010	22	dwell 76, tilt 2	22→13→25	.000566→.001843
A_NCP 3e-4	45011	11	dwell 89	12→0→12	.000721→.002428
A_CFC 1e-3	45010	44	dwell 56	0→8→44	.000648→.000965
A_CFC 1e-3	45011	57	dwell 43	2→14→40	.000631→.001127
A_CFC 1e-3	45012	65	dwell 35	0→34→44	.00047→.001041
A_CFC 3e-4	45010	18	dwell 70, tilt 12	8→0→6	.000887→.002938
A_CFC 3e-4	45011	36	dwell 64	2→0→8	.000889→.002709
A_GRU 1e-3	45010	100	— (dwell 0)	18→100→100	.000168→.000112

D Early-resolution formula and savings

For a per-arm precondition of mean $\geq 90\%$ over 10 seeds, after k seeds with partial sum S the leg is declared FAIL when

$$S + (10 - k) \cdot 100 < 900.$$

Applied to the four legs: A_NCP@3e-4 FAIL at k=2 (S=33; 833 < 900); A_NCP@1e-3 FAIL at k=4 (S=289; 889 < 900); A_CFC@3e-4 FAIL at k=2 (S=54; 854 < 900); A_CFC@1e-3 FAIL at k=3 (S=166; 866 < 900). Total executed: 13 cycles versus 40 for a full n=10 over four legs; one four-seed batch (45016–45019) was never launched. This is arithmetic on the frozen criterion, not a modification of it.

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